

A Tenth Memoir on Quantics

(Continuation from earlier pages, Paper No. 693)

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[The preceding paragraphs (printed pages 354–355) construct from the original quintic $(a, b, c, d, e, f \mid \xi, \eta)^5$ the canonical reduced form and explain the genesis of the capital covariants $A, B, C, \dots, V, W$.]

This is

$$\xi^5 + 5\xi^4\eta + 10\xi^3\eta^2 + 10\xi^2\eta^3 + 5\xi\eta^4 + \eta^5,$$

which is

$$\begin{aligned} &= (1 - b + b^2 - b^3 + b^4 - b^5) \\ &\quad - 2b + 3b^2 + 5b^4 \\ &+ b(1 - 3b - 4b^2) \\ &+ ac(1 - 10b^2) \\ &+ a^2d(1 + 6b^2) \\ &+ a^3e(-4b + 10b^2) \\ &+ af(-5b) \\ &\quad 1 \\ &\quad 1). \end{aligned}$$

$$= (1 \quad ae + 1 \quad a^2d + 1 \quad a^3e + 1 \quad a^4f + 1 \mid \xi, \eta)^5$$

with the subsidiary numerical schedule

$$\begin{array}{cccc} b^2 - 1 & abc - 3 & a^2bd - 4 & a^3be - 5 \\ & b^3 + 2 & ab^2c + 6 & a^2bcd + 10a^2b^2c \text{ (etc.)} \\ & & b^4 - 3 & ab^3c - 10 \\ & & & b^5 + 4 \end{array}$$

The values of a, b, c, d, e, f , considered for a moment as denoting the leading coefficients of the several covariants ultimately represented by these letters respectively, are

$$\begin{array}{cccccc} a & b & c & d & e & f \\ a + 1 & ae + 1 & ac + 1 & ace + 1 & a^2f + 1 + a^2c^2d & 1 \\ & & & & abe + 5 & \\ bd - 4 & b^2 - 1 & ad^2 - 1 & & ac^2d + 2 & abc - 3 \\ c^2 + 3 & & & & ab^2d + 8 & b^3 - 2 \\ & & & -b^2e \cdot 1 & bcd + 2 & bc^2 - 10 \\ & & & bcd + 2 & -c^3 \cdot 1 & \end{array}$$

satisfying, as they should do, the relation

$$f^2 = -a^2d + a^2bc - 4c^3.$$

Hence forming the values of $a^4b \cdot 5c^2$ and $a^3e \cdot 2cf$, it appears that the value of the last-mentioned quintic function is

$$(1, 0, c, f, a^2b - 3c^2, a^3e - 2cf \mid \xi, \eta)^5.$$

Writing herein x, y in place of ξ, η , and now using a, b, c, d, e, f to denote, not the leading coefficients but the covariants themselves (a denoting the original quintic, with ξ, η as facients), we have the form

$$\mathfrak{A} = (1, 0, c, f, a^2b - 3c^2, a^3e - 2cf \mid x, y)^5,$$

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a new quintic, which is the canonical form in question: the covariants hereof (reckoning the quintic itself as a covariant) will be written $A, B, C, \dots, V, W$, and will be spoken of as *capital covariants*.

376. The fundamental property is: Every capital covariant, say I , has for its leading coefficient the corresponding covariant i multiplied by a power of a : and this follows as an immediate consequence of the foregoing genesis of A . The covariant i of the form

$$\frac{1}{n}(a, b, c, d, e, f \mid \xi, \eta)^n$$

has a leading coefficient

$$= \frac{1}{n}(a^2cf - a^2be + \&c.),$$

which, when $a, b, c, d, e, f, \dots, i$ denote leading coefficients, is i multiplied by a power of a : and upon substituting for the quintic the linear transformation thereof

$$(1, 0, c, f, a^2b - 3c^2, a^3e - 2cf \mid \xi, \eta)^5,$$

(observing that, in the transformation ξ, η into $\xi + h\eta, a\eta$, the determinant of substitution is $= a$), the value is still $= i$ multiplied by a power of a ; or using the relation $\mathfrak{a} = a$, say the value is $= i$ multiplied by a power of a . Now the covariant i is the same function of the covariants $a, b, c, d, e, \dots$ that the leading coefficient i is of the leading coefficients a, b, c, d, e, f ; hence, the italic letters now denoting covariants, the leading coefficient still is $= i$ multiplied by a power of a : which is the above-mentioned theorem.

377. To show how the transformation is carried out, consider, for example, the covariant B . This is obtained from the corresponding covariant of $(a, b, c, d, e, f \mid \xi, \eta)^5$, that is,

$$\begin{aligned} & (ae \cdot 1 \quad af \cdot 1 \quad bf \cdot 1 \mid \xi, \eta)^2, \\ & \begin{array}{ccc} bd - 4 & be - 3 & ce - 4 \\ & cd + 1 & \\ c^2 + 1 & & d^2 + 3 \end{array} \end{aligned}$$

by changing the variables, and for the coefficients

$$a, b, c, d, e, f$$

writing

$$1, 0, c, f, a^2b - 3c^2, a^3e - 2cf;$$

thus the coefficients are

First.	Second.	Third.
$1(a^2b - 3c^2)$	$-1(a^3e - 2cf)$	$-4c(a^2b - 3c^2)$
	$+2cf$	
$+3c^2$	$= a^3e$	$+3f^2$
$= a^2b$		$= -4a^2bc + 12c^4$
		$+3(-a^2d + a^2bc - 4c^3)$
		$= a^2(-3ad - bc);$

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and we have thus the expression of B (see the Table No. 97); and similarly for the other capital covariants $C, D, \dots, V, W$: in every case the coefficients are obtained in the standard form, that is, as rational and integral functions of a, b, c, d, e, f , linear as regards f .

378. It will be observed that there is in each case a certain power of a which explicitly divides all the coefficients and is consequently written as an exterior factor: disregarding these exterior factors, the leading coefficients for B, C, D, E, F are b, c, ad, e, f respectively; that for G is $12abd + 4b^2c + e^2$, which must be $=g$ multiplied by a power of a , and (in Table 97) is given as $=a^2g$; similarly, that for H is $6acd + \&c.$, which must be $=h$ multiplied by a power of a , and is given as $=a^4h$: and so in the other cases. The index of a is at once obtained by means of the deg-order, which is in each case inserted at the foot of the coefficient.

For A, B, C, E, F there is no power of a as an interior factor: and for the invariants $G, Q, \dots$ we may imagine the interior factor thrown together with the exterior factor, ($G = a^2g, \&c.$): whence disregarding the exterior factors, we may say that for $A, B, C, E, F, G, Q, \dots$ the standard forms are also “divided” forms.

But take any other covariant, for instance, D : the leading coefficient is ad , having the interior factor a ; and this being so it is found that all the following coefficients will divide by a (the quotients being of course expressible only in terms of the covariants subsequent to f): thus the second coefficient of D is $-bf + ce$, and (5.11) we have $-bf + ce = ai$, or the coefficient divided by a is i ; and so for the other coefficients of D ; or throwing out the factor a , we obtain for D an expression of the form $(d, i, \dots | x, y)^4$, see the Table 98: this is the “divided” form of D : and we have similarly a divided form for every other capital covariant. All that has been required is that each coefficient of the divided form shall be expressed as a rational and integral function of the covariants $a, b, c, \dots, v, w$: and the form is not hereby made definite: to render it so, the coefficient must be expressed in the *segregate* form. But there is frequently the disadvantage that we thus introduce fractions; for instance, the last coefficient of D is $ci + df$, where to get rid of the congregate term df we have (6.12), $3df = -al + 2ct$, and the segregate form of the coefficient is

$$= \frac{1}{3}al + \frac{1}{3}ct.$$

379. We have in regard to the canonical form, a differential operator which is analogous to the two differential operators $xd_y \{xd_y\}, yd_x \{yd_x\}$ considered in the Introductory Memoir (1854), [139]. Let δ denote a differentiation in regard to the constants under the conditions

$$\begin{aligned}\delta a &= 0, \\ \delta b &= e, \\ \delta c &= 3f, \\ \delta d &= \frac{1}{a}(-bf + ce), \quad (= i), \\ \delta e &= -6ad - 10bc, \\ \delta f &= -2a^2b - 18c^2,\end{aligned}$$

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which (as is at once verified) are consistent with the fundamental relation

$$f^2 = -a^2d + a^2bc - 4c^3;$$

then it is easy to verify that

$$\left(x \frac{d}{dy} - 4cy \frac{d}{dx} - \delta\right) A = 0;$$

and this being so, any other covariant whatever, expressed in the like standard form, is reduced to zero by the operator

$$x \frac{d}{dy} - 4cy \frac{d}{dx} - \delta;$$

and we have thus the means of calculating the covariant when the leading coefficient is known.

Thus, considering the covariant B , the expression of which has just been obtained, $= (B_0, B_1, B_2 \mid x, y)^2$, suppose: the equation to be satisfied is

$$\begin{aligned} & x(B_1x + 2B_2y - 4cy(2B_0x + B_1y) \\ & - x^2\delta B_0 - xy\delta B_1 - y^2\delta B_2 = 0, \end{aligned}$$

viz. we have

$$\begin{aligned} B_1 - \delta B_0 &= 0, \\ 2B_2 - 8cB_0 - \delta B_1 &= 0, \\ -4cB_1 - \delta B_2 &= 0; \end{aligned}$$

which (omitting, as we may do, the outside factor a^2) are satisfied by the foregoing values $B_0, B_1, B_2 = b, e, -3ad - bc$. And if we assume only $B_0 = b$, then the first equation gives at once the value $B_1 = e$, the second equation then gives $B_2 = -3ad - bc$; and the third equation is satisfied identically, viz. the equation is

$$-4ce + 8(3ad + bc) = 0,$$

that is,

$$-4ce = -4ce = 0,$$

$$\begin{aligned} & + c\delta b + c \cdot e \\ & + b\delta e + b \cdot 3f \\ & + 3a\delta d + 3(-bf + ce) \end{aligned}$$

which is right.

Of course every invariant must be reduced to zero by the operation δ : thus we have, see the Table No. 97,

$$a^2g = 12abd + 4b^2c + 1e^2,$$

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and thence

$$\begin{aligned} a^2\delta g &= (12ad + 8bc)\delta b &= (12ad + 8bc)e \\ &+ 4b^2 \cdot \delta c &+ 4b^2 \cdot 3f \\ &+ 12ab \cdot \delta d &+ 12b(-bf + ce) \\ &+ 2e \cdot \delta e &+ 2e(-6ad - 10bc), \end{aligned}$$

with row-totals (in the columns ade, b^2f, bce):

	<i>ade</i>	<i>b²f</i>	<i>bce</i>	Sum
	+12		+8	
		+12		
	-12		+12	
	-12		-20	
	0	0	0	

which is = 0, as it should be.

380. As already remarked, the leading coefficients of H, I, J , &c., are each of them equal to a power of a multiplied by the corresponding covariant $h, i, j, \dots$; hence, supposing these leading coefficients, or, what is the same thing, the standard expressions of the covariants $h, i, j, \dots, v, w$ to be known, we can calculate the values of $\delta h, \delta i, \delta j, \dots, \delta v, \delta w$ ($= 0$, since w is an invariant): and the operation δ , instead of being applicable only to the forms containing a, b, c, d, e, f , becomes applicable to forms containing any of the covariants. The values of $\delta a, \delta b, \dots, \delta v, \delta w$ can, it is clear, be expressed in terms of segregates; and this is obviously the proper form: but for $\delta r, \delta t$, and δv , for which the segregate forms are fractional, I have given also forms with integer coefficients. The entire series is

Deg-order.

$$2.8 \quad \delta a = 0,$$

$$3.5 \quad \delta b = e,$$

$$3.9 \quad \delta c = 3f,$$

$$4.6$$

$$4.8 \quad \delta d = i,$$

$$4.12$$

$$5.3 \quad \delta e = -6ad - 10bc,$$

$$5.7$$

$$5.9 \quad \delta f = -2a^2b - 18c^2,$$

$$6.4$$

$$6.6 \quad \delta g = 0,$$

$$6.10$$

$$7.5 \quad \delta h = 2je - 4i,$$

$$7.7$$

$$8.4 \quad \delta i = -2ab^2 + 2ah - 18cd,$$

$$8.8$$

$$9.3 \quad \delta j = -n,$$

$$9.5$$

$$10.6 \quad \delta k = 2aj + 6b^2j - 9bh + 3c^2,$$

$$12.4$$

$$13.3 \quad \delta l = -3abd - 7b^2c + 7ch,$$

$$14.4$$

$$\delta m = -bk - p,$$

$$19.3$$

$$\delta n = 4cj,$$

$$\delta o = \dots$$

$$\delta p = b^2g + 6bm - 6dj - gh,$$

$$8ahj - 5adg - 10b^4 + 15b^2c^2 - 5bcg + 10cm,$$

$$\delta q = 0,$$

$$\delta r = \frac{1}{3}(aq + 6b^2j - 5bdg - jh), \quad = 2b^2j - 2bdg - 6dm,$$

$$\delta s = 2agj + 2f^2g + 3b^2m + 21bdj - 4bgh + 2c^2f^2 - 3cq,$$

$$\delta t = \frac{1}{3}(bgm + \frac{4}{3}bf^2 - 3dgj - kq), \quad = -b^2q + hq + 6m^2,$$

$$\delta u = 0,$$

$$\delta v = \frac{1}{3}(-5bgr - 10bjo + 5gjk - 12js - 9nq), \quad = -6dt - 6mr + nq,$$

$$\delta w = 0.$$

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It is obvious that for every covariant whatever written in the denumerate form $(I_0, I_1, \dots | x, y)^n$, the second coefficient is equal to the first coefficient operated upon by δ ; so that the foregoing formulae give, in fact, the second coefficients of the several covariants.

381. It is worth noticing how very much the formulae of Table No. 97 simplify themselves, if one of the covariants b, c, d, e vanishes, in particular, if b vanishes. Suppose $b = 0$; writing also (although this makes but little difference) $a = 1$, we have

$$\begin{aligned} a &= 1, \\ b &= 0, \\ c &= c, \\ d &= d, \\ e &= e, \\ f^2 &= -d - 4c^3, \\ g &= e^2, \\ h &= 6cd + ef, \\ i &= ce, \\ j &= 9d^2 + c^4, \\ k &= 3de, \\ l &= -3df + 2c^2e, \\ m &= 9cd^2 + 3def - c^4e, \\ n &= -6cde - e^2f, \\ o &= 9d^2e + c^2e^3, \\ p &= +9d^2f + 12c^3de - c^2ef, \\ q &= 54cd^3 - 27d^2ef + 18c^3de^2 + c^6f, \\ r &= 9cd^2e + 3de^3f - c^5e^2, \\ s &= -27d^3f + 54c^2d^2e + 9cde^3f - 2c^5e^3, \\ t &= -81d^4f - 6d^2f^3 + 216c^3d^2e + 54cd^3f - 24c^4de^2 - c^7ef, \\ u &= -27d^4 - 18cd^2e^2 - 4d^2e^2f + c^6de^4, \\ v &= 81d^4ef - 6d^4f^3 + 216c^3d^2e^2 + 54cd^3ef - 24c^4de^4 - 7c^7e^2f, \\ w &\text{ (not calculated).} \end{aligned}$$

These values are very convenient for the verification of syzygies, &c. Take, for instance, the before-mentioned relation $\delta v = -6dt - 6mr + nq$, that is, if $V = (V_0, V_1 | x, y)$, then $V_1 = -6dt - 6mr + nq$: calculating the three products on the right-hand side, observing

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that f^2 when it occurs is to be replaced by its value $-d - 4c^3$, and taking their sum, the figures are as follows:

	$-6dt$	$-6mr$	$+nq$	Sum
d^5f	+486			+486
	+36			
d^3f^3		+54	-27	+63
	-1296			
$c^3d^3e^2$	-324	-486	+324	-1458
cd^4e^2f		-324	+216	-432
			+1	+1
			-216	
$c^4d^2e^4$	+144	+324	-24	+252
	+6	+36		
c^6de^4f			+4	+18
		-6		
			-2	

where the last column is, in fact, what V_1 becomes on writing therein $a = 1, b = 0$. The verification would not of course apply to terms which contain b ; thus, (13.3), a derived syzygy is $\delta l = 6t + mo$; and the foregoing values give, as they should do, $\delta l = mo$: we might for the verification of most of the terms in b use values $a, b, c, d, e, f^2 = -d$: 1, $b, 0, d, e$, the only failure would be for terms containing bc .

Table No. 97.

Covariants of A, in the af- or standard forms; W is not given.

[The Table presents the capital covariants $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V$ each expressed as a polynomial in x, y whose coefficients are listed in vertical columns of monomials in the lowercase letters a, b, c, d, e, f , with numerical multipliers and deg-order labels at the foot of each column. The structure is reproduced below.]

Covariant A: the original quintic. Coefficients of $(x, y)^5$:

$$A = (1 ; c + 10 ; f + 10, c^2 - 15 ; a^2b + 5, cf - 2 ; a^3e + 1 \mid x, y)^5$$

with deg-orders 0.0, 1.3, 2.6, 3.9, 4.12, 5.15.

Covariant B $= a^2$ of order 2 in (x, y) :

$$B = a^2(b + 1 ; e + 1 ; ad - 3, bc - 1 \mid x, y)^2$$

with deg-orders 2.2, 3.5, 4.8.

Covariant C: order 6,

$$C = (c + 1 ; f + 1 ; a^2b + 3, c^2 - 15 ; a^2b + 1, cf - 10 ; a^2d + 6, e^2 + 15 ; af^2 \cdot 1, a^2ce + 3, a^2cf + 1, a^2cf + 1 ; a^3b^2 - 1, a^2cd + 2, a^2b^2c + 4, a^2ef + 1, a^3f^2 - 1 \mid x, y)^6$$

deg-orders 2.6, 3.9, 4.12, 5.15, 6.18, 7.21, 8.24.

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Covariant D $= a \cdot$ order 4:

$$D = a(ad + 1 ; bf - 1, ce + 1 ; a^2b^2 - 1, acd + 3, a^2bc^2 + 4, ef + 1 ; \dots \mid x, y)^4$$

deg-orders 4.8, 5.11, 6.14, 7.17.

Covariant $E = a^2 \cdot$ order 5:

$$E = a^2(e + 1 ; ad \cdot 6, a^2bc \cdot 10 ; bf - 12, ce + 2 ; a^2b^2 \cdot 8, acd \cdot 36, a^2bc^2 + 12, a^2ef \cdot 2 ; \dots | x, y)^5$$

further columns: $a^2be \cdot 5$, $adf - 1$, $a^2bcf \cdot 4$, $a^2c^2e + 2$; $a^3bd \cdot 6$, $a^2b^2c^2 \cdot 2$, $a^2e^2 \cdot 1$, $ac^3d + 18$, $a^2b^2c + 6$, $a^2cef + 2$.
deg-orders 3.5, 4.8, 5.11, 6.14, 7.17, 8.20.

Covariant F : order 4 (very long — shown across columns):

$$F = (f + 1 ; a^2b + 2, a^3c - 18 ; a^2e + 1, a^3c - 36, ac^2 + 168 ; a^2d + 34, a^2bc \cdot 42, a^3cf - 126 ; \dots | x, y)^4$$

further columns: $a^2bf \cdot 40$, $a^2ce + 5$, $a^2bc^2 + 134$, $a^2ef \cdot 5$, $a^2c^3 \cdot 252$; $a^4b^2 \cdot 16$, $a^2cd + 6$, $a^2bcf + 8$, $a^2c^2e + 65$, $a^2c^2f \cdot 84$; $\dots$; $a^2b^2d \cdot 12$, $a^2bcd + 6$, $\dots$, $a^2c^4 \cdot 2$.

Deg-orders 3.9, 4.12, 5.15, 6.18, 7.21, 8.24, 9.27, 10.30, 11.33, 12.36.

Covariant $G = a^2g$ (invariant): the single coefficient

$$G = a^2(12abd + 4b^2c + e^2) \quad (\text{deg-order } 6.10).$$

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Covariant $H = a^4h$: order 4:

$$\begin{aligned} H = a^4 & (acd + 3, a^2bc^2 + 4, a^2ef + 1; a^2be + 2, adf + 12, a^2bcf \cdot 8, a^2c^2e \cdot 8; \\ & a^2bc + 4, a^2e^2 + 1, ac^2d \cdot 36, a^2bc^2 \cdot 24, a^2ef \cdot 6; \\ & a^2de + 2, a^2bcf + 4, a^2bce \cdot 6, ac^2d \cdot 12, a^2bcf \cdot 8, a^2c^3e + 8; \\ & a^2b^2 + 2, a^2c^2 + 6, a^2b^2d \cdot 2, a^2bcd \cdot 8, a^2bef \cdot 3, a^2ce^2 + 1, ac^3d \cdot 6, a^2bd^2 + 4, a^2c^2ef + 1 \\ & = a^4h \quad | x, y)^4 \end{aligned}$$

deg-orders 6.14, 7.17, 8.20, 9.23, 10.26.

Covariant $I = a^2i$: order 5:

$$\begin{aligned} I = a^2 & (bf \cdot 1, ce + 1; a^2b^2 \cdot 2, acd \cdot 6, a^2b^2c + 8, a^2ef \cdot 2; adf \cdot 15, a^2bef + 5, a^2c^2e \cdot 5; \\ & a^2bd \cdot 20, ac^2d + 8, a^2bce \cdot 5, acdf + 30, a^2bcf + 5, a^2c^3e \cdot 5; \\ & a^2c^2e \cdot 5; a^2b^2 + 2, a^2d^2 \cdot 12, a^2b^2cd \cdot 2, a^2b^2c^2 \cdot 6, a^2ce^2 \cdot 2, ac^3d \cdot 30, a^2bc^4 \cdot 8, a^2c^3ef \cdot 2; \\ & a^4b^4 + 1, a^2b^2cf + 3, a^2cde \cdot 5, a^2b^2cf + 1, a^2bc^2e \cdot 5, a^2c^2df \cdot 8, a^2bc^4 \cdot 1, a^2c^2e + 1 \quad | x, y)^5 \end{aligned}$$

$= a^2i$, deg-orders 5.11, 6.14, 7.17, 8.20, 9.23, 10.26, 11.29.

Covariant $J = a^4j$: order 2:

$$\begin{aligned} J = a^4 & (a^2b^2 + 1, a^2d^2 + 9, a^2b^2c^2 \cdot 4, a^2b^2f \cdot 2, a^2ce^2 + 1; \\ & a^2b^2c \cdot 1, abdf \cdot 6, a^2cde + 6, a^2b^2cf \cdot 4, a^2bc^2e + 8, a^2ef \cdot 1 \quad | x, y)^2 = a^4j \end{aligned}$$

deg-orders 8.16, 9.19.

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Covariant $K = a^4k$: order 4:

$$\begin{aligned} K = a^4(& ade + 3, a^2bcf + 2, a^2bce \cdot 2; a^2b^2 + 4, a^2d^2 \cdot 18, abcd \cdot 18, a^2b^2c^2 \cdot 16, a^2bef \cdot 5, a^2ce^2 + 1; \\ & a^2bh + 1, abdf + 6, a^2b^2c^2 \cdot 2, a^2bc^2e \cdot 2, a^2ef \cdot 1; \\ & a^2b^2d + 6, a^2b^2c + 2, a^2cd^2 \cdot 18, a^2b^2c^2 \cdot 30, a^2def \cdot 9, a^2b^2c^2 \cdot 8, a^2bcef \cdot 5, a^2c^4 + 3 \quad | x, y)^4 \\ & = a^4k \end{aligned}$$

deg-orders 7.13, 8.16, 9.19, 10.22.

Covariant $L = a^2l$: order 4 (deg-orders up to 14.38):

$$L = a^2(adf \cdot 3; a^2bd \cdot 3, a^2b^2c \cdot 7, ac^2d + 42, a^2bc^3 + 28, a^2cef + 7; \dots | x, y)^4$$

deg-orders 7.17, 8.20, 9.23, 10.26, 11.29, 12.32, 13.35, 14.38.

[The remaining covariants $M, N, O, P, Q, R, S, T, U, V$ in this Table are reproduced in compact symbolic form in Table No. 98 below (divided and (mostly) segregate form), with their full numerical coefficient schedules; only the leading-coefficient identities are retained here.]

Covariant $M = a^4m$: order 4, deg-orders 10.22, 11.25, 12.28.

Covariant $N = a^2n$: order 4, deg-orders 9.19, 10.22, 11.25, 12.28, 13.31.

Covariant $O = a^4o$: order 4, deg-orders 11.21, 12.24.

Covariant $P = a^2p$: order 4, deg-orders 11.25, 12.28, 13.31, 14.34, 15.37, 16.40.

Covariant $Q = a^6q$ (invariant), deg-order 14.30.

Covariant $R = a^6r$: order 2, deg-orders 13.27, 14.30, 15.33.

Covariant $S = a^4s$: order 2, deg-order 15.33.

Covariant $T = a^8t$: order 2, deg-orders 19.41, 20.44, 21.45.

Covariant $U = a^{10}u$ (invariant), deg-order 12.0 (in this normalisation).

Covariant $V = a^{12}v$: deg-orders 22.46, 23.49.

[The intermediate detail (precise integer multipliers of each monomial term in every coefficient of every covariant) occupies six tightly-set printed pages of Table No. 97 in the original; the principal use of these schedules is verification of syzygies, and they are not used in the running argument that follows. The scan-image of the table is preserved alongside this transcription as a separate IA-linked artefact.]

[Vol. X, printed page 366; PDF page 386. Table No. 97 (continued).]

[Schedule of covariant N (order 4), deg-orders 9.19, 10.22, 11.25, 12.28, 13.31, with leading monomials a^2b^4 , a^2b^2e , a^2bde , a^2b^4 , a^2b^4e respectively; full coefficient lists in scan.]

[Schedule of covariant O (order 4, $= a^4o$), deg-orders 11.21, 12.24, with leading monomials $a^2b^2e + 3$ and $a^2b^2d \cdot 6$; full coefficient lists in scan.]

[Schedule of covariant P (order 4): leading columns $P = a^2(a^2bde \cdot 2, a^2d^2 \cdot 1; a^2b^4 \cdot \dots; a^2b^2e \cdot 5, a^2d^2e \cdot 9; a^2b^4d \cdot 8, a^2d^3 \cdot 72; a^2b^2de + 6, a^2b^4f + 5, \dots; a^2b^2 + 2, a^2b^2d^2 + 12, \dots \mid x, y)^4$, deg-orders 11.25, 12.28, 13.31, 14.34, 15.37, 16.40.]

[Vol. X, printed page 367; PDF page 387. Table No. 97 (continued).]

Covariant $Q = a^6q$ (invariant), deg-order 14.30:

$$Q = a^6(a^2b^3 \cdot 2, a^2b^2d^2 + 18, a^2b^2cd + 22, a^2bde^2 + 3, a^2cd^2 \cdot 54, a^2b^2c^2 + 12, \dots, a^2cef^2 + 1).$$

Covariant $R = a^6r$ (order 2), deg-orders 13.27, 14.30, 15.33:

$$R = a^6(r_0; r_1; r_2 \mid x, y)^2$$

with r_0, r_1, r_2 given as in Table 98 below (divided segregate form).

Covariant $S = a^4s$ (order 2), deg-orders 15.33, 16.36, 17.39, 18.42:

$$S = a^4(s_0; s_1; s_2; s_3 \mid x, y)^2.$$

[Vol. X, printed page 368; PDF page 388. Table No. 97 (continued).]

Covariant $T = a^8 t$ (order 2), deg-orders 19.41, 20.44, 21.45:

$$T = a^8 (t_0; t_1; t_2 \mid x, y)^2, \quad = a^8 t.$$

Covariant $U = a^{10} u$ (invariant): the single coefficient is the polynomial in a, b, c, d, e, f of deg-order 21.45; for the simplified $b = 0$ value see Art. 381 above.

[Vol. X, printed page 369; PDF page 389. Table No. 97 (concluded).]

Covariant $V = a^{12}v$ (deg-orders 22.46, 23.49):

$$V = a^{12}(v_0; v_1 \mid x, y) = a^{12}v.$$

The coefficient v_1 contains the monomial schedule listed on the right-hand of the original page; only the principal columns are reproduced in compact form in Table 98.

Table No. 98.

Covariants of A, divided and (except as to a few coefficients) segregate.

[Vol. X, printed page 370; PDF page 390.]

A and B as given in Table 97 were divided and segregate.

C was divided but not segregate: the divided and segregate form is

$$C = (c + 1; f + 3, a^2b + 3, c^2 \cdot 15; a^2b + 1, cf \cdot 10, c^3 + 15; a^2d + 6, a^2bc \cdot 3; \\ a^2i + 3, a^2cf + 3; a^4b^2 \cdot 1, a^2h + 1, a^2cd \cdot 4, a^2c^3 \cdot 1 \mid x, y)^6$$

deg-orders 2.6, 3.9, 4.12, 5.15, 6.18, 7.21, 8.24.

D divided and segregate is

$$D = a(d + 1; i + 1; ab^2 \cdot 1, ah + 1, a^2cd \cdot 3; al \cdot 1, act \cdot 1 \mid x, y)^4$$

deg-orders 3.3, 4.6, 5.9, 6.12. An integer non-segregate form of the fractional coefficient is

$$\frac{ci \cdot 1}{df + 1}$$

E was divided but not segregate: the divided and segregate form is

$$E = (e + 1; ad \cdot 6, a^2bc \cdot 10; ai + 12, a^2ce \cdot 10; a^2b^2 \cdot 8, ah \cdot 2, acd \cdot 24, a^2bc^2 + 20; \dots \mid x, y)^5$$

further columns: $a^2be \cdot 5, a^2b^2 \cdot 1, ac^2 \cdot 12, a^2bc + 5; a^2c^2 \cdot 1, a^2bd + 6, a^2b^2c + 2, a^2cd^2 \cdot 2, a^2c^3d + 6, a^2bc^2 \cdot 2.$

deg-orders 3.6, 4.8, 5.11, 6.14, 7.17, 8.20.

[Vol. X, printed page 371; PDF page 391. Table No. 98 (continued).]

F was divided but not segregate: the divided and segregate form is

$$F = (f + 1 ; a^2b + 2, a^2c \cdot 18 ; ah + 1, a^2c \cdot 36 ; a^2d + 34, a^2bc \cdot 42, a^2cf + 168 ; \dots | x, y)^4$$

remaining columns: $a^2l + 40, a^2ce \cdot 35 ; a^4b^2 \cdot 16, a^2h \cdot 5, a^2cf \cdot 16 ; a^2be \cdot 21, a^2l \cdot 8, a^2cf \cdot 252, a^2ce + 189 ; \dots ; a^4b^2 \cdot 2, a^4bh \cdot 3, a^4cf \cdot 16, a^4cn \cdot 5, a^4cd + 16, a^4bc^2 + 4, a^4c^4 \cdot 2.$

Deg-orders 3.9, 4.12, 5.15, 6.18, 7.21, 8.24, 9.27, 10.30, 11.33, 12.36.

where for an integer non-segregate value of the fractional coefficient, see the original form of F .

G as an invariant was divided and segregate, $G = a^2g$. (Deg-order 4.0.)

H divided and segregate is

$$H = a^4(h + 1 ; be + 2, i \cdot 4 ; a^2g + 1, abd \cdot 12, a^2ch \cdot 6 ; a^2fk \cdot 2, abi \cdot 8, a^2bce \cdot 6, a^2cj + 12 ; a^2j + 2, a^2h + 4, a^2bh \cdot 5, a^2cp + 1, abcd + 12, a^2c^2h + 3 | x, y)^4$$

deg-orders 4.4, 5.7, 6.10, 7.13, 8.16. Where the fractional coefficients are

$$\begin{array}{l|l} abe + 2 & a^2f^2 + 2 \\ a^2bf + 4 & a^2d^2 + 6 \\ a^2bce \cdot 6 & abcd \cdot 2 \\ a^2cl + 4 & a^2b^2c \cdot 8 \\ & a^2bef \cdot 3 \\ & a^2c^2h + 1 \\ & a^2ce^2 + 1 \end{array}$$

[Vol. X, printed page 372; PDF page 392. Table No. 98 (continued).]

I divided and segregate is

$$I = a^2(i + 1 ; ab^2 \cdot 2, ah + 2, a^2cd \cdot 18 ; al + 5, a^2cl \cdot 15 ; a^2bd \cdot 20, a^2c^2d + 60 ; \dots | x, y)^5$$

remaining columns: $a^2b^2k \cdot 5, a^2bi \cdot 25, aci \cdot 30, a^2d^2 + 45 ; a^2ck \cdot 4, a^2b^2 + 10, a^2bh \cdot 8, a^2cg + 4, a^2bcd \cdot 6, a^2bc^2 \cdot 18, a^2c^2f \cdot 6, a^2c^2d \cdot 54 ; \dots ; a^2c^2t \cdot 3.$

deg-orders 4.6, 5.9, 6.12, 7.15, 8.18, 9.21, 10.24. Where the fractional coefficients are

$$\begin{array}{l|l|l} a^2de \cdot 5 & a^2b^2 + 2 & a^2b^2e + 1 \\ abf + 5 & a^2d^2 \cdot 12 & a^2bdf + 3 \\ a^2bce \cdot 5 & a^2bcd \cdot 2 & a^2cde \cdot 5 \\ a^2c^2i \cdot 5 & ab^2c \cdot 6 & ab^2c + 1 \\ a^2cdf + 30 & a^2c^2h \cdot 2 & a^2bc^2e \cdot 5 \\ & a^2ce^2 \cdot 2 & a^2cf \cdot 3 \\ & a^2c^2i \cdot 18 & a^2c^2i + 1 \\ & & a^2cdf \cdot 3 \end{array}$$

J divided and segregate is

$$J = a^4(j, = a^4(x, y)^0).$$

(Deg-order 5.6 and 6.4.)

K divided and segregate is

$$K = a^4(k + 1 ; aj \cdot 2, a^2b^2 + 6, a^2bh \cdot 9, a^2cg + 3 ; an + 1, a^2ck \cdot 3 ; a^2m \cdot 3, acj + 1, a^2b^2c \cdot 2, a^2bch + 3 | x, y)^4$$

deg-orders 5.3, 6.6, 7.9, 8.12.

[Vol. X, printed page 373; PDF page 393. Table No. 98 (continued).]

L divided and (as to first six coefficients) segregate is

$$\frac{1}{3}L = a^2(l + 1 ; abd \cdot a^2k \cdot 4 ; a^2j \cdot 13, a^2c \cdot 3, a^4m \cdot 1, a^2bde \cdot 7, a^4b^2 \cdot 2 ; \dots | x, y)^4$$

further columns: $abi + 1$, am ., a^4bf ., $a^4b^2c \cdot 7$, $a^4b^2d + 13$, $a^4bd^3 + 3$; $a^2ch + 7$, $a^4cl \cdot 21$, $a^4bh \cdot 5$, $a^4ck \cdot 20$, $a^4cj + 3$, $a^4b^2c^2 + 14$, $a^4b^2cd + 10$, $a^4de^2 + 2$; . . .

deg-orders 5.7, 6.10, 7.13, 8.16, 9.19, 10.22, 11.25, 12.28. The last two coefficients have not been reduced to the segregate form.

M divided and segregate is

$$M = a^4(m + 1 ; bk \cdot 1, l^2 \cdot 1 ; abj \cdot 11, adg + 1, a^2cm \cdot 1 | x, y)^4$$

deg-orders 6.2, 7.5, 8.8.

[Vol. X, printed page 374; PDF page 394. Table No. 98 (continued).]

N divided and segregate is

$$N = a^4(n+1; cj+4; ap \cdot 6, cn \cdot 6; a^2bj \cdot 4, a^2dg+4, ab^4+8, a^2b^2h \cdot 12, a^2eg+4, a^2dm \cdot 8, a^2c^2j \cdot 4; a^4o+1, a^4bn \cdot 1, a^4cp+2, a^4c^2j+1 \mid x, y)^4$$

deg-orders 6.4, 7.7, 8.10, 9.13, 10.16.

O divided and segregate is

$$O = a^4(o+1; b^2g+1, bm+6, c^2 \cdot 6, ch \cdot 1 \mid x, y)^4$$

deg-orders 7.1, 8.4.

P divided and (as to first three coefficients) segregate is

$$P = a^2(p+1; abj+8, a^2dg \cdot 5, a^2bl \cdot 14, a^2b^2c+15, a^2bcg \cdot 5, a^2cm+10; \dots \mid x, y)^4$$

further columns: a^2o+7 , $ab^4 \cdot 2$, $o^2 \cdot 14$, $a^2b^2ch \cdot 8$; a^2bm+8 , a^2b^2d+9 , $a^4bcj+13$, $a^4bdh \cdot 9$, $a^4c^4 \cdot 81$, $a^4b^2c \cdot 5$, $a^4b^2ch \cdot 8, \dots$

and: $a^2b^2k \cdot 3$, $a^2bp \cdot 3$, a^2cj^2+2 , ab^2de+3 , a^2bl+3 , a^2bh+8 , $a^4dh \cdot 27$, $a^4cdh \cdot 3, \dots$; $a^4b^2+; \dots$

deg-orders 7.8, 8.8, 9.11, 10.14, 11.17, 12.20. The last three coefficients have not been reduced to the segregate form.

[Vol. X, printed page 375; PDF page 395. Table No. 98 (continued).]

Q as an invariant was divided and segregate, $Q = a^6q$ (deg-order 8.0).

R divided and segregate is

$$R = a^6(r + 1 ; aq + 1, a^2b^2j + 6, a^2bdg \cdot 5, a^2hj \cdot 1 ; abo \cdot 1, a^2gk \cdot 1, a^2s + 3, a^2cr \cdot 3 \mid x, y)^2$$

deg-orders 8.2, 9.5, 10.8. Where the fractional coefficients are

$$\begin{array}{l|l} b^2j + 2 & bdk + 3 \\ bdg \cdot 2 & bej + 1 \\ dm \cdot 6 & cr + 1 \\ & d^2j \cdot 1 \\ & dp + 3 \end{array}$$

S divided and (as to the first three coefficients) segregate is

$$S = a^4(s + 1 ; agj \cdot 2, a^2b^2g + 2, a^2b^2m + 3, a^2bdj + 21, a^2b^2h \cdot 4, a^2cg^2 + 2, a^2cq \cdot 3 ; \dots \mid x, y)^2$$

further columns: $abr \cdot 1, a^2do \cdot 1 ; a^2bq + 4, a^2b^2j + 4, a^2b^2k \cdot 4, a^2bdm \cdot 31, a^2d^2j \cdot 3, a^4b^2 + 4, a^4b^2cj + 16, a^4bd^2h \cdot 24, a^4dcn + 4, \dots$

deg-orders 9.3, 10.6, 11.9, 12.12. But the last coefficient is neither segregate nor integer.

[Vol. X, printed page 376; PDF page 396. Table No. 98 (concluded).]

T divided and segregate is

$$T = a^8(t + 1 ; bgm + 1, k^2 + 4, dgj \cdot 3, kq \cdot 1 \mid x, y)^2$$

deg-orders 11.1, 12.4. Where the fractional coefficient is

$$\frac{b^2q \cdot 1}{hq + 1} \\ m^2 + 6$$

U as an invariant was divided and segregate, $U = a^{10}u$ (deg-order 12.0).

V divided and segregate is

$$V = a^{12}(v + 1 ; bgr \cdot 5, bjo \cdot 10, gjk + 5, js \cdot 12, nq \cdot 9 \mid x, y)^6$$

deg-orders 13.1, 14.4. Where the fractional coefficient is

$$\frac{dt \cdot 6}{mr \cdot 6} \\ nq + 1$$

W as an invariant was divided and segregate, $W = a^{18}w$ (deg-order 18.0).

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Derivatives. Art. Nos. 382 to 384, and Tables Nos. 99 and 100.

382. I call to mind that any two covariants a , b , the same or different, give rise to a set of derivatives $(a, b)^1$, $(a, b)^2$, $(a, b)^3$, &c., or, as I propose to write them, $ab1$, $ab2$, $ab3$, &c., viz.

$$\begin{aligned} ab1 &= d_x a \cdot d_y b - d_y a \cdot d_x b, \\ ab2 &= d_x^2 a \cdot d_y^2 b - 2 d_x d_y a \cdot d_x d_y b + d_y^2 a \cdot d_x^2 b, \\ ab3 &= d_x^3 a \cdot d_y^3 b - 3 d_x^2 d_y a \cdot d_x d_y^2 b + 3 d_x d_y^2 a \cdot d_x^2 d_y b - d_y^3 a \cdot d_x^3 b, \\ &\text{\&c.} \end{aligned}$$

or, as these are symbolically written,

$$ab1 = 12 a_1 b_2, \quad ab2 = 12^2 a_1 b_2, \quad ab3 = 12^3 a_1 b_2, \quad \text{\&c.};$$

where

$$12 = \frac{d}{d\xi_1} \cdot \frac{d}{d\eta_2} - \frac{d}{d\eta_1} \cdot \frac{d}{d\xi_2};$$

the differentiations $\frac{d}{d\xi_1}$, $\frac{d}{d\eta_1}$ applying to the a , and the $\frac{d}{d\xi_2}$, $\frac{d}{d\eta_2}$ applying to the b , but the suffixes being ultimately omitted: hence if θ be the index of derivation, the derivative is thus a linear function of the differential coefficients of the order θ of the two covariants a and b respectively: and we have the general property that any such derivative, if not identically vanishing, is a covariant. If the a and the b are one and the same covariant, then obviously every odd derivative is $= 0$; so that in this case the only derivatives to be considered are the even derivatives $aa2$, $aa4$, &c. Moreover, if the index of derivation θ exceeds the order of either of the component covariants, then also the derivative is $= 0$: in particular, neither of the covariants must be an invariant. The degree of the derivative is evidently equal to the sum of the degrees of the component covariants; the order is equal to the sum of the orders less twice the index of derivation.

383. It was by means of the theory of derivatives that Gordan proved (for a binary quantic of any order) that the number of covariants was finite, and, in the particular case of the quintic, established the system of the 23 covariants. Starting from the quantic itself a , then the system of derivatives $aa2$, $aa4$, &c., must include among itself all the covariants of the second degree, and if the entire system of these is, suppose, b , c , &c., then the derivatives $ab1$, $ab2$, &c., $ac1$, $ac2$, &c., must include among them all the covariants of the third degree, and so on for the higher degrees: and in this way, limiting by general reasoning the number of the independent covariants of each degree obtained by the successive steps, the foregoing conclusion is arrived at. But returning to the quintic, and supposing the system of the 23 covariants established, then knowing the deg-order of a derivative we know that it must be a linear function of the segregates of that deg-order; and we thus confirm, *à posteriori*, the results of the derivation theory. I annex the following Table No. 99, showing all the derivatives which present themselves, and for each of them the

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covariants as well congregate as segregate of the same deg-order: the congregates are distinguished each by two prefixed dots, *.bf*, &c. No further explanation of the arrangement is, I think, required. We see from the table in what manner the different covariants present themselves in connexion with the derivation-theory. Thus starting with the quintic itself *a*, we have the two derivatives *aa4*, *aa2*, which are in fact the covariants of the second degree (deg-orders 2.2 and 2.6 respectively) *b* and *c*. For the third degree we have the derivatives *ab2*, *ab1*, *ac5*, *ac4*, *ac3*, *ac2*, *ac1*: the deg-order of *ac5* is 3.1, and there being no covariants of this deg-order, *ac5* must, it is clear, vanish identically: *ab2* and *ac4* are each of them of the deg-order 3.3, but for this deg-order we have only the covariant *d*, and hence *ab2* and *ac4* must be each of them a numerical multiple of *d*; similarly, deg-order 3.5, *ab1* and *ac3* must be each of them a numerical multiple of *e*; deg-order 3.7, *ac2* must be a numerical multiple of *ab*; and deg-order 3.9, *ac1* must be a numerical multiple of *f*: the 7 derivatives, which *primâ facie* might give, each of them, a covariant of the third degree, thus give in fact only the 3 covariants *d*, *e*, *f*; and in order to show according to the theory of derivations that this is so, it is necessary to prove 1°, that *ac5* = 0; 2°, that *ac4* and *ab2* differ only by a numerical factor; 3°, that *ab1* and *ac3* differ only by a numerical factor; 4°, that *ac1* is a numerical multiple of *ab*: which being so, we have the 3 new covariants. The table shows that

for degrees 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24
No. of derivatives = 2, 7, 19, 29, 41, 46, 52, 46, 44, 35, 26, 19, 17, 12, 13, 6, 6, 3, 3, 1, 1, 0, 1

so that the whole number of derivatives is 429, giving the 22 covariants *b, c, . . . , w*. While it is very remarkable that (by general reasoning, as already mentioned, and with a very small amount of calculation) Gordan should have been able in effect to show this, the great excess of the number of derivatives over that of the covariants seems a reason why the derivations ought not to be made a basis of the theory.

It is to be remarked that we may consider derivatives *pq1*, *pq2*, &c., where *p, q* instead of being simple covariants are powers or products of covariants, but that these may be made to depend upon the derivatives formed with the simple covariants. (As to this see my paper “On the Derivatives of Three Binary Quantics,” *Quart. Math. Journal*, t. xv. (1877), pp. 157–168, [681].)

Table No. 99 (Index Table of Derivatives).

Deg. 2 (2 derivs.)

Ord.	2	4	6
<i>aa</i>		<i>b</i>	<i>c</i>

Deg. 3 (7 derivs.)

Ord.	1	3	5	7	9
<i>ab</i>		<i>d</i> (= 4)	<i>e</i> (= 2)		
<i>ac</i>	5	4	3	2	1

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Table No. 99 (continued).

Deg. 4 (19 derivs.) — covariants present: $b^2, i, ad, ae, a^2b, h, bc, cd, ce, \dots$

Ord.	2	4	6	8	10	12
ad	3	2	1			
ae	5	4	3	2	1	
af		5	4	3	2	1
bb	2					
bc		2	1			
ce	6	4		2		

Deg. 5 (29 derivs.) — covariants: $j, k, ag, be, ab^2, ai, a^2d, bd, l, ah, ..bf, abc, cd, ce$

Ord.	1	3	5	7	9	11	13
ah	4	3	2	1			
ai	5	4	3	2	1		
bd	2	1					
be		2	1				
$c-e$		2	1				
cd	3	2	1				
ce	4	3	2	1			
cf	6	5	4	3	2	1	

Deg. 6 (41 derivs.) — covariants include $bg, n, aj, a^2f, ak, av, abe, a^2b^2, a^2h, ci, acd, bh, ab^2c, bc^2, ag, ..df, ..d^2, ch, ..e^2, ..ef$

Ord.	2	4	6	8	10	12	14
aj				1			
ak		3	2	1			
al	3	4	3	2	1		
bh	2	1					
bi		2	1				
ch	4	3	2	1			
ci	6	5	4	3	2	1	
dd	2						
de	3	2	1				
df			3	2	1		
ee	4		2				
ef		5	4	3	2	1	
ff	8		6		4		2

[Vol. X, printed page 380; PDF page 400. Table No. 99 (continued).]

Deg. 7 (46 derivs.) — covariants present at heads of columns: $hj, hk, abg, an, a^2j, a^2k, dg, eg, am, ..b^2e, ab^3, abi, p, bM, bl, ..cde, ..ab^3, abh, ck, acg, ..ad^2, bcd, bce, ..dh, ..di, ..eh, cl, ..ei, ..fh, fg, ..by$ etc.

Ord.	1	3	5	7	9	11	13
am		2	1				
an	4	3	2	1			
bj	1						
bk	2	1					
bl			2	1			
cj			1				
ck		3	2	1			
cl	6	5	4	3	2	1	
dh	3	2	1				
di		3	2	1			
eh	4	3	2	1			
ei	5	4	3	2	1		
fh			4	3	2	1	
fi		6	5	4	3	2	1

[The remainder of Table No. 99 (covering degrees 8 through 24 of derivatives, with the same column-by-deg-order layout) continues across the following printed pages and forms a single large index; it is not reproduced here. Total: 429 derivatives, yielding the system of 22 covariants $b, c, \dots, w$ of the binary quintic.]

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